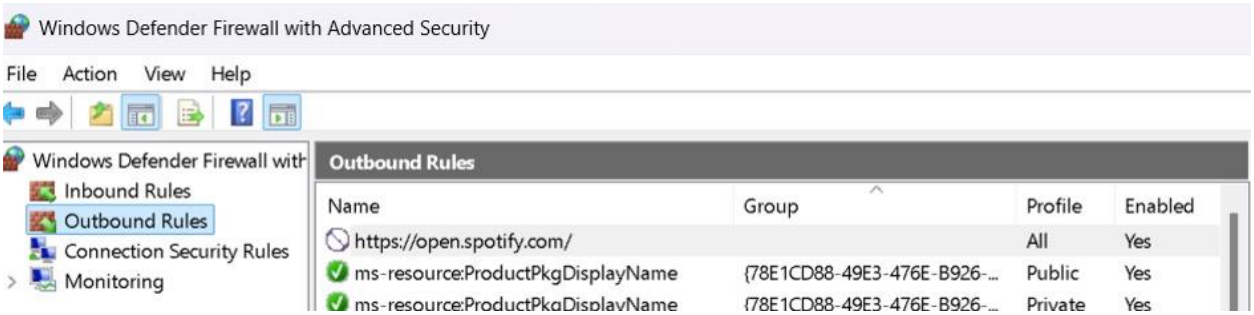
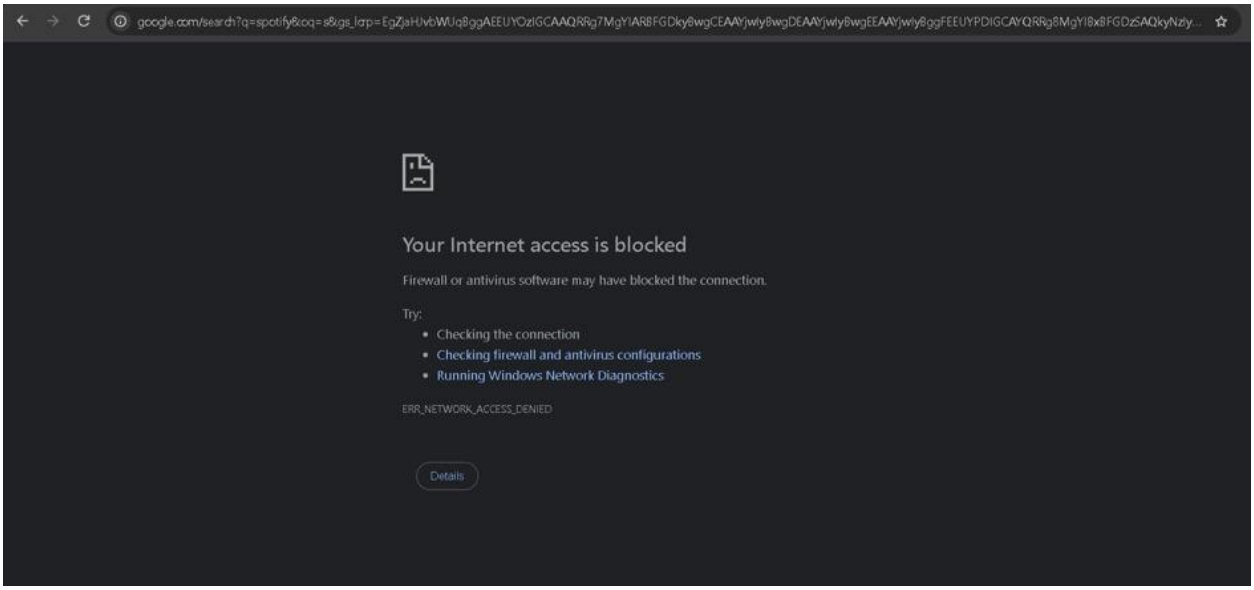
Day 4: Firewall Basics Theory :Firewall types and use cases (hardware and software firewalls).Practical :Configure basic rules in a software firewall (e.g., Windows Defender).Task: Document how to block and allow specific traffic using a firewall.

Tasks:-

1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Ans.:-



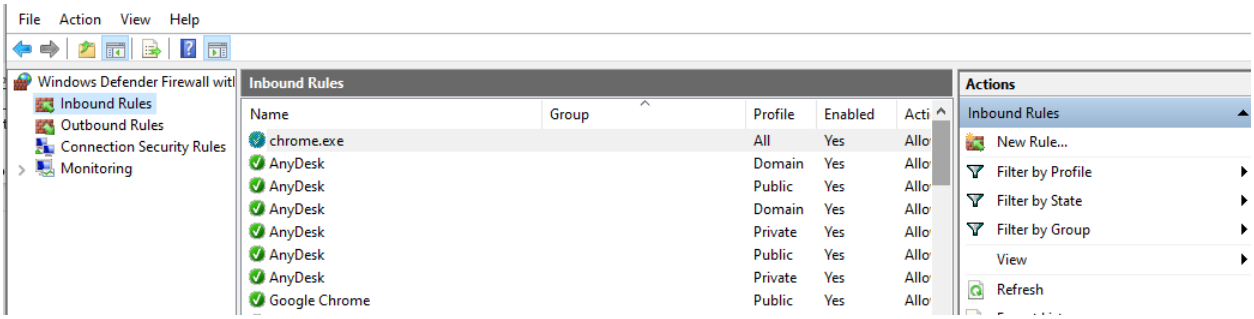


2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Ans.:-

Steps:-

1. Press Windows + R, type wf.msc, and press Enter.
2. In Windows Defender Firewall with Advanced Security, click Inbound Rules.
3. Click New Rule... in the right pane.
4. Select Program and click Next.
5. Choose This program path, then click Browse.
6. Select the Chrome executable, usually located at:
    - C:\Program Files\Google\Chrome\Application\chrome.exe
7. Click Next.
8. Select Allow the connection, then click Next.
9. Keep Domain, Private, and Public selected, then click Next.
10. Enter the rule name:
    - Allow Chrome HTTPS
11. Click Finish.



**3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.**

Ans.:-

Steps:-

1. Press Windows + R, type wf.msc, and press Enter to open Windows Defender Firewall with Advanced Security.
2. Click Inbound Rules in the left pane.
3. Click New Rule... in the right pane.
4. Select Custom and click Next.
5. Under Program, leave All programs selected and click Next.
6. Under Protocol and Ports, leave Protocol type as Any and click Next.
7. In the Scope tab:
    - Under Which remote IP addresses does this rule apply to?
    - Select These IP addresses.
    - Click Add....
    - Enter the IP address:

192.168.1.4

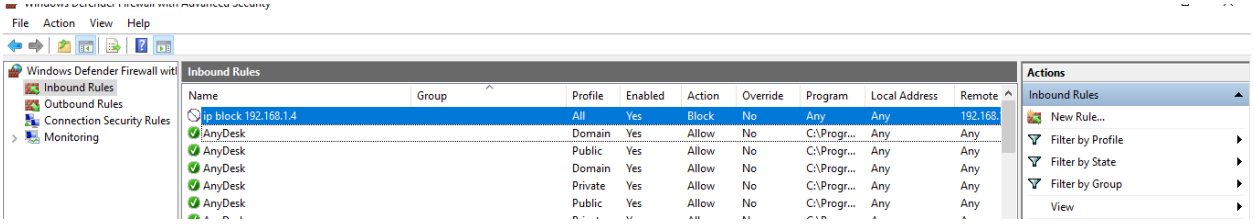
    - Click OK, then Next.
8. Select Block the connection and click Next.
9. Select all three profiles:
    - ✓ Domain
    - ✓ Private
    - ✓ Public
10. Click Next.
11. Enter the rule name:

Block IP 192.168.1.8

12. (Optional) Add a description:

Blocks all incoming traffic from IP address 192.168.1.4

13. Click Finish.



4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Ans.:-

Feature	Hardware Firewall	Software Firewall (Windows Defender Firewall)
Definition	A dedicated physical device that filters network traffic before it reaches computers.	A security program installed on a computer that monitors and controls network traffic.
Installation	Installed between the modem/router and the internal network.	Installed directly on a PC, laptop, or server.
Coverage	Protects the entire network and all connected devices.	Protects only the device on which it is installed.
Performance	Uses its own hardware, so it has little or no impact on computer performance.	Uses the computer's CPU and memory, which may slightly affect performance.
Security Level	Provides strong network-level protection against external attacks and unauthorized access.	Protects against unwanted connections and controls application access on the local device.

Use Cases:-

Hardware Firewall:-

Example**:** A company with 300 employees installs a hardware firewall between its ISP connection and office network. It protects all employee computers, printers, servers, and Wi-Fi devices from cyberattacks while allowing only approved network traffic.

Software Firewall (Windows Defender Firewall):-

Example**:** A student uses a Windows laptop at home and on public Wi-Fi. Windows Defender Firewall blocks unauthorized applications and hackers from accessing the laptop while allowing safe internet access for browsers, email, and other trusted applications.